

LAPORAN PRAKTIKUM
SISTEM OPERASI
MODUL 9: SYSCALL



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BAB I

PENDAHULUAN

1.1 Latar Belakang

Sistem operasi merupakan perangkat lunak inti yang mengatur sumber daya komputer, seperti proses, memori, dan perangkat masukan-keluaran. Salah satu mekanisme penting pada sistem operasi adalah system call atau syscall. Syscall menjadi penghubung antara program pengguna dan kernel agar layanan kernel dapat digunakan secara terkontrol.

Pada praktikum Modul 9 digunakan Xinu sebagai media pembelajaran. Praktikum ini berfokus pada pembuatan syscall baru, perbaikan syscall chprio, dan pengujian perubahan prioritas proses melalui perintah ps dan uptime.

1.2 Tujuan Praktikum

- Mengimplementasikan syscall baru pada Xinu.
- Memperbaiki syscall chprio dengan validasi input PID dan prioritas.
- Mengubah xsh_uptime.c agar dapat memanggil syscall.
- Menguji hasil menggunakan ps, uptime, dan ps kembali.

1.3 Rumusan Masalah

- Bagaimana menambahkan syscall baru pada Xinu?
- Bagaimana melakukan validasi input pada chprio?
- Bagaimana membuktikan prioritas PID 5 berubah atau tetap sesuai skenario pengujian?

BAB II

LANDASAN TEORI

2.1 Xinu

Xinu adalah sistem operasi sederhana yang digunakan untuk mempelajari struktur kernel, proses, shell command, dan prioritas proses secara langsung.

2.2 System Call

System call adalah mekanisme pemanggilan layanan kernel. Pada praktikum ini dibuat syscall baru chname dan digunakan syscall chprio untuk mengganti prioritas proses.

2.3 Syscall chprio

chprio digunakan untuk mengganti prioritas proses. Validasi yang ditambahkan adalah PID harus berada pada rentang valid dan prioritas harus bernilai positif.

```
if (pid < 0 || pid >= NPROC) {
    restore(mask);
    return SYSERR;
}

if (newprio <= 0) {
    restore(mask);
    return SYSERR;
}
```

BAB III

METODOLOGI PRAKTIKUM

3.1 Alat dan Bahan

- VirtualBox
- Development-system Xinu
- Backend Xinu
- Terminal Linux, gedit, make, dan minicom
- Source code Xinu

3.2 Alur Praktikum

- Membuat syscall chname.
- Menambahkan prototype chname.
- Memperbaiki chprio.c.
- Mengubah xsh_uptime.c.
- Compile dengan make clean dan make.
- Run dengan sudo minicom dan reset backend.
- Uji dengan ps, uptime, dan ps kembali.

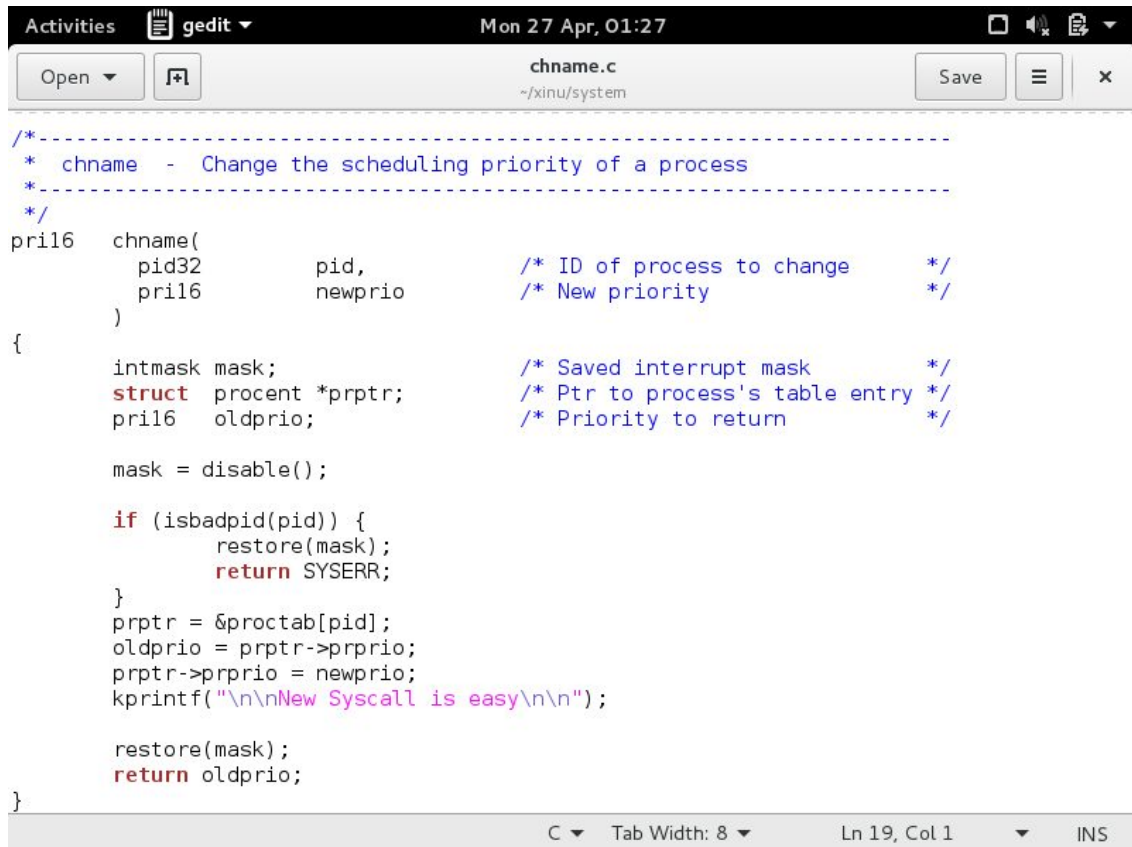
```
cd ~/xinu/compile
make clean
make
sudo minicom
```

BAB IV

IMPLEMENTASI DAN HASIL PRAKTIKUM

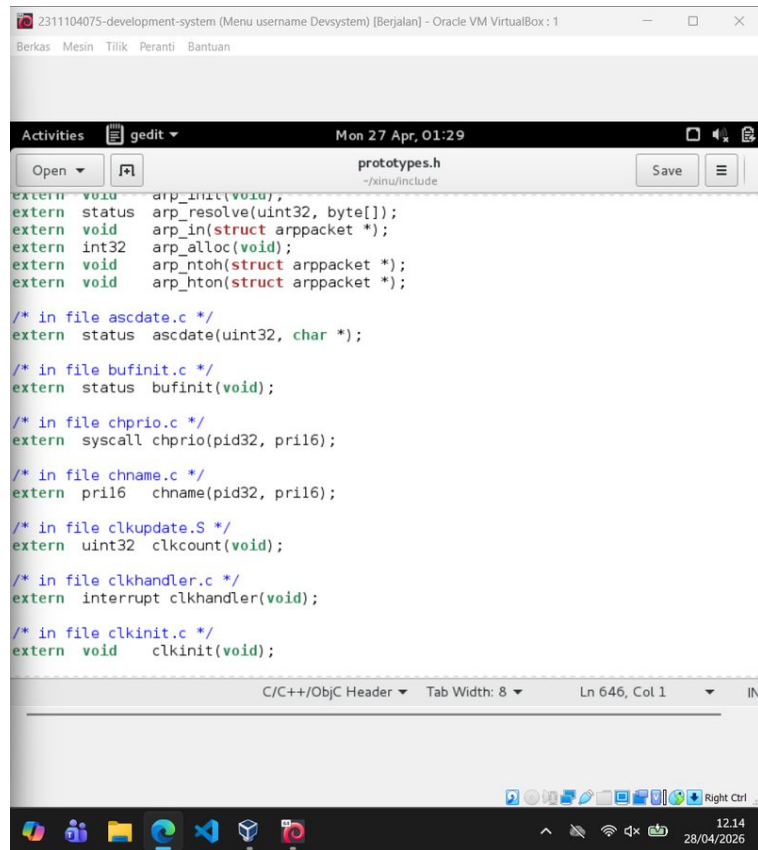
4.1 Pembuatan Syscall Baru chname

Syscall baru dibuat pada file chname.c. Fungsi ini menerima PID dan prioritas baru, memvalidasi PID, mengambil data proses dari proctab, mengubah prioritas, lalu mencetak pesan New Syscall is easy.



```
/*-----  
* chname - Change the scheduling priority of a process  
*-----  
*/  
pril6 chname(  
    pid32      pid,          /* ID of process to change */  
    pril6      newprio       /* New priority             */  
)  
{  
    intmask mask;           /* Saved interrupt mask     */  
    struct procent *prptr;   /* Ptr to process's table entry */  
    pril6 oldprio;          /* Priority to return        */  
  
    mask = disable();  
  
    if (isbadpid(pid)) {  
        restore(mask);  
        return SYSERR;  
    }  
    prptr = &proctab[pid];  
    oldprio = prptr->prprio;  
    prptr->prprio = newprio;  
    kprintf("\n\nNew Syscall is easy\n\n");  
  
    restore(mask);  
    return oldprio;  
}
```

Gambar 4.1 Source code syscall baru chname.c



```
extern void arp_init(void);
extern status arp_resolve(uint32, byte[]);
extern void arp_in(struct arppacket *);
extern int32 arp_alloc(void);
extern void arp_ntoh(struct arppacket *);
extern void arp_hton(struct arppacket *);

/* in file ascdatetime.c */
extern status ascdatetime(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

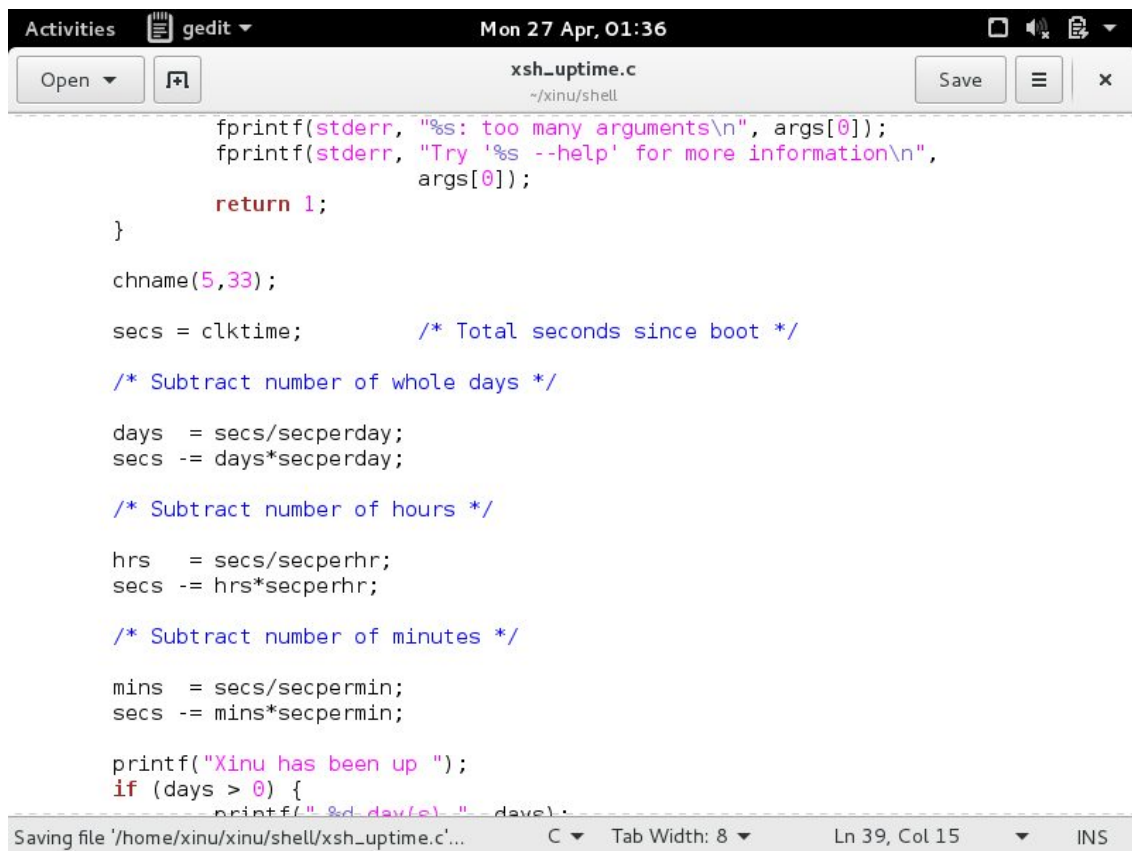
/* in file cname.c */
extern pri16 cname(pid32, pri16);

/* in file clkupdate.S */
extern uint32 clkcount(void);

/* in file clkhandler.c */
extern interrupt clkhandler(void);

/* in file clkinit.c */
extern void clkinit(void);
```

Gambar 4.2 Penambahan prototype cname pada prototypes.h



```
fprintf(stderr, "%s: too many arguments\n", args[0]);
fprintf(stderr, "Try '%s --help' for more information\n",
        args[0]);
return 1;
}

cname(5,33);

secs = clktime; /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

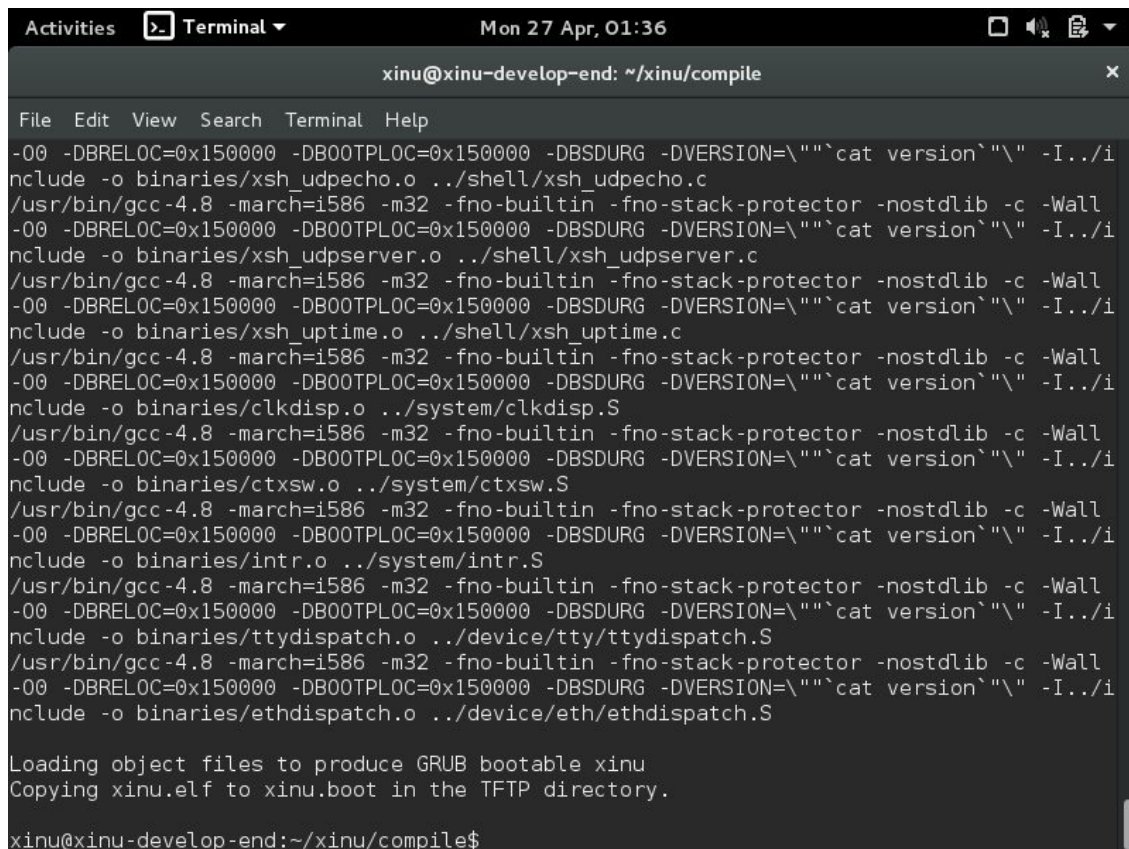
hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf("%d day(s) ", days);
}
```

Gambar 4.3 Pemanggilan cname(5,33) pada xsh_uptime.c



```
Activities Terminal Mon 27 Apr, 01:36
xinu@xinu-develop-end: ~/xinu/compile

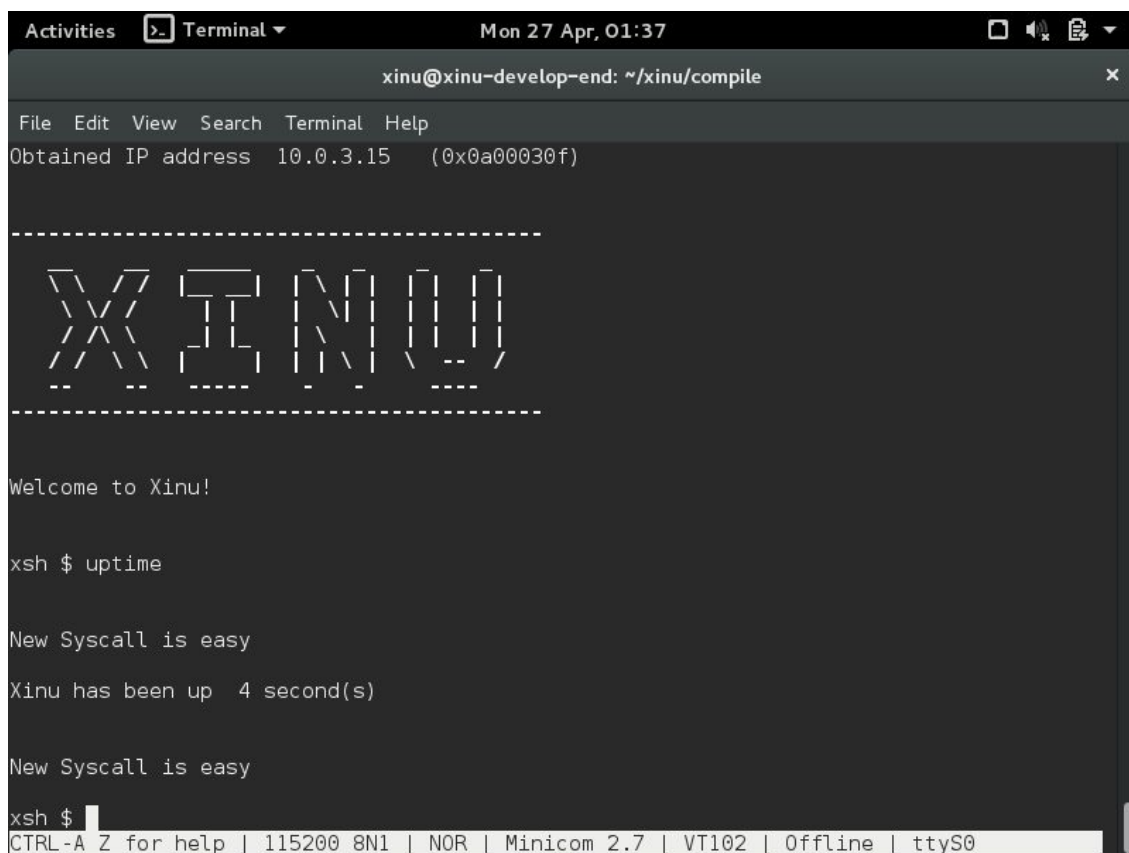
File Edit View Search Terminal Help

-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/xsh_udpsvr.o ../shell/xsh_udpsvr.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/clkdisp.o ../system/clkdisp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$
```

Gambar 4.4 Proses compile syscall baru berhasil



```
Activities Terminal Mon 27 Apr, 01:37
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

Obtained IP address 10.0.3.15 (0x0a00030f)

-----
XINU
-----

Welcome to Xinu!

xsh $ uptime

New Syscall is easy

Xinu has been up 4 second(s)

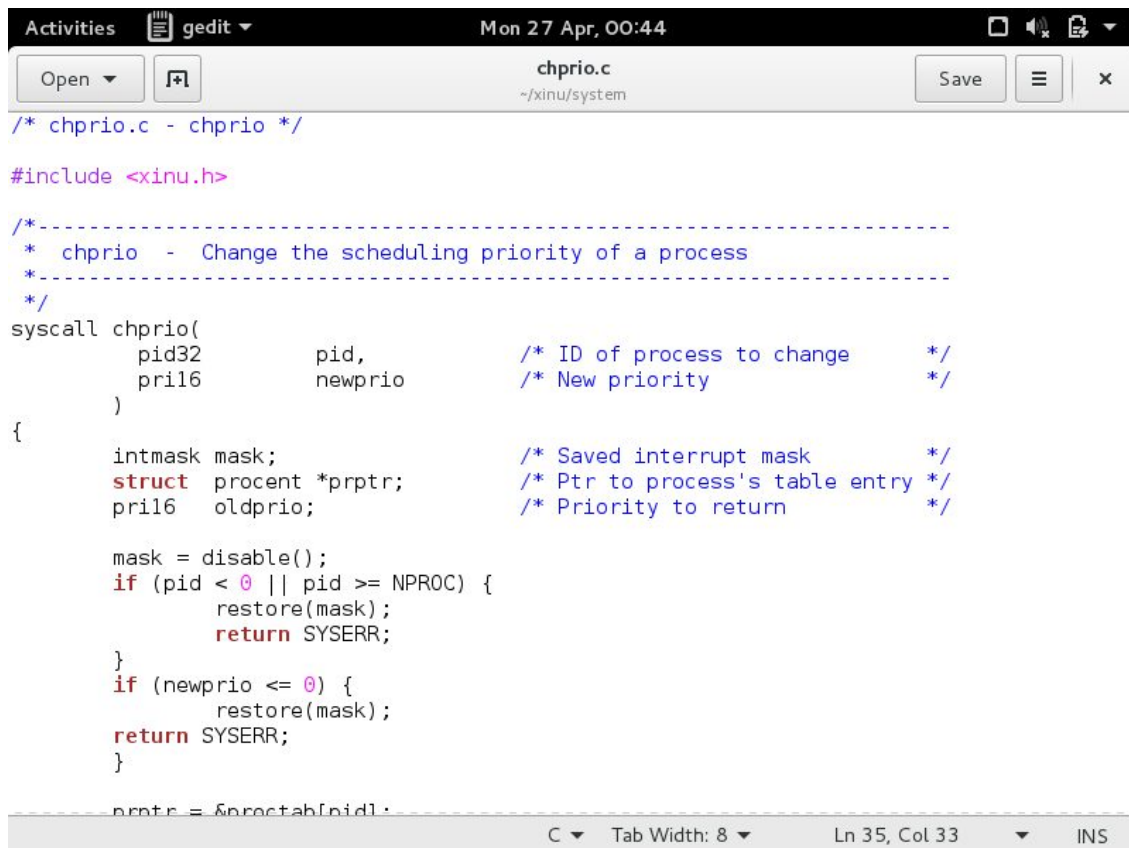
New Syscall is easy

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

Gambar 4.5 Output New Syscall is easy ketika uptime dijalankan

4.2 Perbaikan Syscall chprio

chprio.c diperbaiki dengan validasi PID dan prioritas. Jika PID tidak valid atau prioritas tidak positif, fungsi mengembalikan SYSERR dan prioritas proses tidak diubah.



```

/* chprio.c - chprio */

#include <xinu.h>

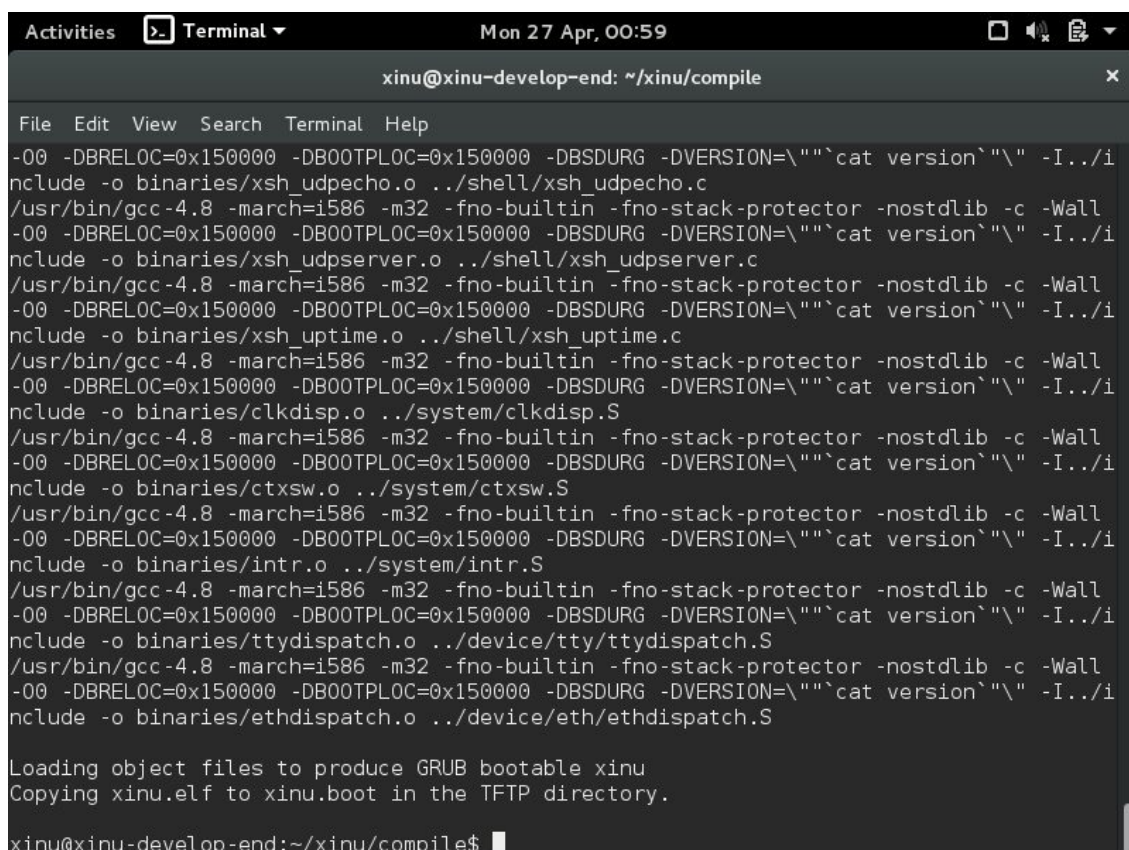
/*-----
 * chprio - Change the scheduling priority of a process
 *-----
 */
syscall chprio(
    pid32      pid,          /* ID of process to change */
    pri16      newprio       /* New priority */
)
{
    intmask mask;           /* Saved interrupt mask */
    struct procent *prptr;   /* Ptr to process's table entry */
    pri16 oldprio;           /* Priority to return */

    mask = disable();
    if (pid < 0 || pid >= NPROC) {
        restore(mask);
        return SYSERR;
    }
    if (newprio <= 0) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];

```

Gambar 4.6 Validasi PID dan prioritas pada chprio.c



```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/xsh_udpserver.o ../shell/xsh_udpserver.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""cat version"" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

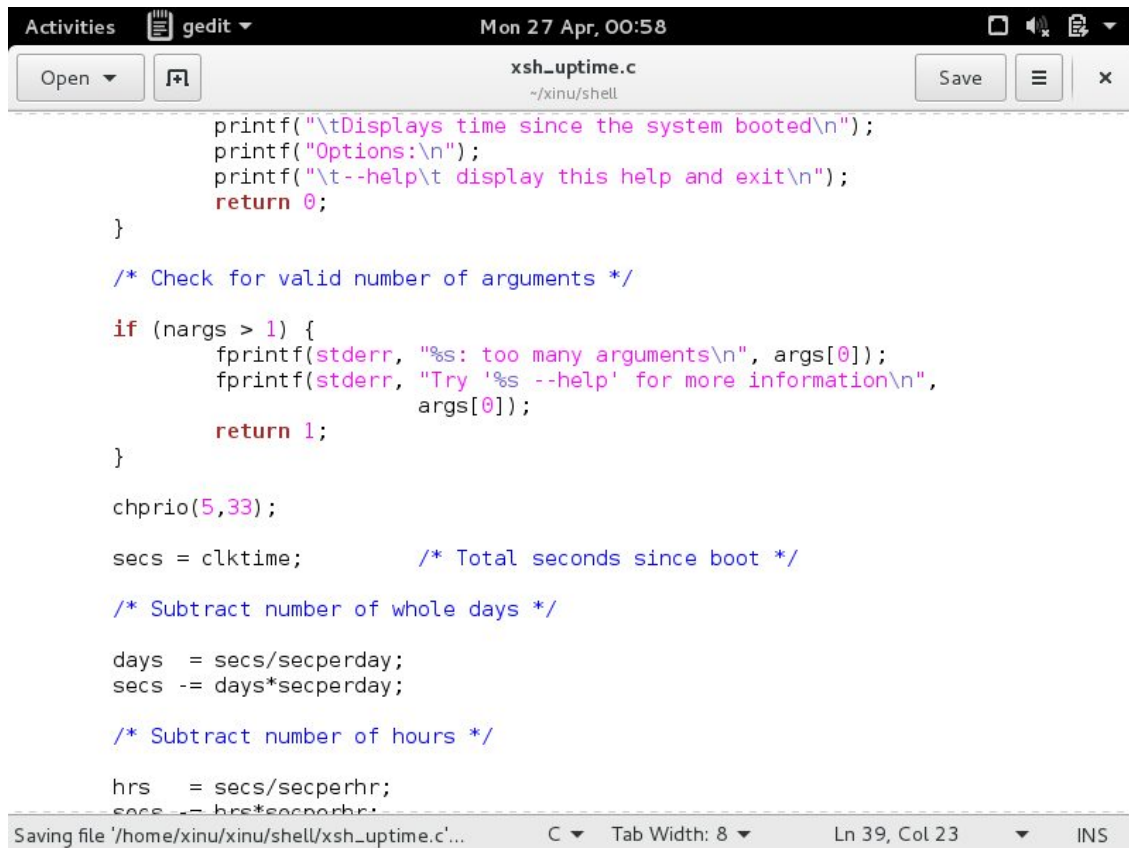
Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$

```

Gambar 4.7 Compile chprio.c berhasil

4.3 Pengujian chprio(5,33)



```
printf("\tDisplays time since the system booted\n");
printf("Options:\n");
printf("\t--help\t display this help and exit\n");
return 0;
}

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

chprio(5,33);

secs = clktime;      /* Total seconds since boot */

/* Subtract number of whole days */

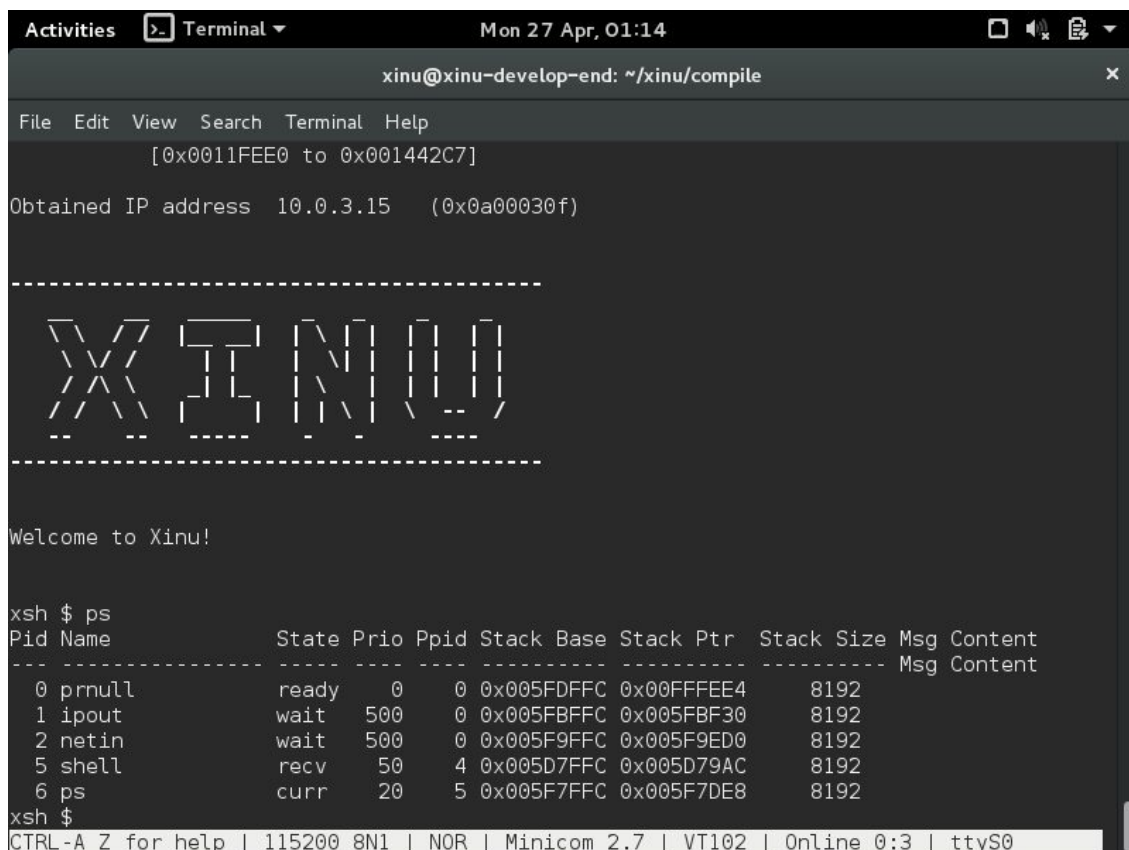
days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;
```

Saving file '/home/xinu/xinu/shell/xsh_uptime.c'... C Tab Width: 8 Ln 39, Col 23 INS

Gambar 4.8 Penambahan chprio(5,33) pada xsh_uptime.c



```
File Edit View Search Terminal Help
[0x0011FEE0 to 0x001442C7]
Obtained IP address 10.0.3.15 (0x0a00030f)

-----
XINU
-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 pnull           ready  0    0 0x005FDFFC 0x00FFFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
5 shell           recv   50    4 0x005D7FFC 0x005D79AC 8192
6 ps              curr   20    5 0x005F7FFC 0x005F7DE8 8192
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:3 | ttyS0
```

Gambar 4.9 Xinu berhasil menampilkan prompt xsh \$

```

xinu@xinu-develop-end: ~/xinu/compile
Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
5 shell           recv   50   4 0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5 0x005F7FFC 0x005F7DE8 8192

xsh $ uptime
Xinu has been up  4 minute(s) 32 second(s)

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE84 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192
5 shell           recv   33   4 0x005D7FFC 0x005D79AC 8192
8 ps              curr   20   5 0x005F7FFC 0x005F7DE8 8192

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:4 | ttyS0

```

Gambar 4.10 Hasil ps - uptime - ps: PID 5 berubah dari 50 menjadi 33

4.4 Pengujian chprio(5,-3)

Pengujian ini memastikan prioritas negatif ditolak. Hasilnya PID 5 tetap memiliki prioritas 50.

```

xsh_uptime.c
~/xinu/shell

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up ");
if (days > 0) {
    printf(" %d day(s) ", days);
}

if (hrs > 0) {
    printf(" %d hour(s) ", hrs);
}

if (mins > 0) {
    printf(" %d minute(s) ", mins);
}

if (secs > 0) {
    printf(" %d second(s) ", secs);
}
printf("\n");
chprio(5,-3);
return 0;
}

```

Gambar 4.11 Pengujian chprio(5,-3)


```

Activities  ▾ Terminal ▾ Mon 27 Apr, 01:18
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_udpsvr.o ../shell/xsh_udpsvr.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DB00TPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$

```

Gambar 4.12 Compile pengujian prioritas negatif berhasil

```

Activities  ▾ Terminal ▾ Mon 27 Apr, 01:20
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

// \ \ | | | \ | \ -- /
-- -- ---- - - - ----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0  0x005FDFFC 0x00FFFE4  8192
1 ipout           wait   500  0  0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0  0x005F9FFC 0x005F9ED0 8192
5 shell           recv   50   4  0x005D7FFC 0x005D79AC 8192
6 ps              curr   20   5  0x005F7FFC 0x005F7FC4 8192

xsh $ uptime
Xinu has been up 19 second(s)

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0  0x005FDFFC 0x00FFFEF0 8192
1 ipout           wait   500  0  0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0  0x005F9FFC 0x005F9ED0 8192
5 shell           recv   50   4  0x005D7FFC 0x005D79AC 8192
8 ps              curr   20   5  0x005F7FFC 0x005F7FC4 8192

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```

Gambar 4.13 Hasil pengujian prioritas negatif: PID 5 tetap 50

4.5 Pengujian chprio(3000,3)

Pengujian ini memastikan PID tidak valid ditolak. Hasilnya PID 5 tetap memiliki prioritas 50.

```

printf("\t--help\t display this help and exit\n");
return 0;
}

/* Check for valid number of arguments */

if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

chprio(3000,3);

secs = clktime;          /* Total seconds since boot */

/* Subtract number of whole days */

days = secs/secperday;
secs -= days*secperday;

/* Subtract number of hours */

hrs = secs/secperhr;
secs -= hrs*secperhr;

/* Subtract number of minutes */

```

Gambar 4.14 Pengujian chprio(3000,3)

```

File Edit View Search Terminal Help
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_udpecho.o ../shell/xsh_udpecho.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_udpserver.o ../shell/xsh_udpserver.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/xsh_uptime.o ../shell/xsh_uptime.c
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/clkdsp.o ../system/clkdsp.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION=""`cat version`"\ -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S

Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$

```

Gambar 4.15 Compile pengujian PID tidak valid berhasil

```

xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192
5 shell           recv  50   4 0x005D7FFC 0x005D79AC 8192
6 ps              curr  20   5 0x005F7FFC 0x005F7DE8 8192

xsh $ uptime
Xinu has been up 11 second(s)

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192
5 shell           recv  50   4 0x005D7FFC 0x005D79AC 8192
8 ps              curr  20   5 0x005F7FFC 0x005F7FC4 8192

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```

Gambar 4.16 Hasil pengujian PID tidak valid: PID 5 tetap 50

4.6 Ringkasan Hasil Pengujian

No	Pengujian	Hasil Aktual	Status
1	chname	New Syscall is easy muncul	Berhasil
2	chprio(5,33)	PID 5 berubah 50 -> 33	Berhasil
3	chprio(5,-3)	PID 5 tetap 50	Berhasil
4	chprio(3000,3)	PID 5 tetap 50	Berhasil

BAB V

ANALISIS DAN PEMBAHASAN

5.1 Analisis Syscall Baru

Output New Syscall is easy membuktikan bahwa syscall chname berhasil dibuat, dipanggil dari xsh_uptime.c, dan masuk ke alur eksekusi Xinu.

5.2 Analisis Perubahan Prioritas

Pada pengujian chprio(5,33), prioritas PID 5 berubah dari 50 menjadi 33 setelah uptime dijalankan. Hal ini membuktikan bahwa perubahan prioritas pada proctab berhasil dilakukan.

5.3 Analisis Validasi

Pada pengujian chprio(5,-3) dan chprio(3000,3), prioritas PID 5 tetap 50. Ini membuktikan bahwa validasi prioritas positif dan validasi PID telah berjalan benar.

5.4 Kendala

Sistem sempat menampilkan program produser-konsumer dari modul sebelumnya. Penyelesaiannya adalah mengembalikan main.c agar menjalankan shell Xinu.



```
Activities gedit Mon 27 Apr, 01:08
Open *main.c Save
~/xinu/system

/*
NAMA : GANES GEMI PUTRA
NIM : 2311104075 */

/*main.c - main*/
#include <xinu.h>

process main(void)
{
    recvclr();

    resume(create(shell, 8192, 50, "shell", 1, CONSOLE));

    return OK;
}
```

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Gambar 5.1 main.c dikembalikan untuk menjalankan shell Xinu

BAB VI

PENUTUP

6.1 Kesimpulan

- Syscall baru berhasil dibuat dan diuji.
- chprio berhasil diperbaiki dengan validasi PID dan prioritas.
- chprio(5,33) berhasil mengubah prioritas PID 5 dari 50 menjadi 33.
- chprio(5,-3) dan chprio(3000,3) berhasil ditolak sehingga prioritas tetap.

6.2 Saran

Pada praktikum berikutnya, praktikan perlu memastikan file yang diedit benar, menyimpan semua perubahan sebelum kompilasi, dan menjalankan pengujian hanya pada prompt xsh \$, bukan terminal Linux.